

Image Quality Equations for Focused Transducer in Circular Photoacoustic Computed Tomography

Soheil Hakakzadeh
Department of Electrical Engineering
Sharif University of Technology
Tehran, Iran
soheil.hakakzadeh@ee.sharif.edu

Zahra Kavehvasht
Department of Electrical Engineering
Sharif University of Technology
Tehran, Iran
kavehvasht@sharif.edu

Abstract— In this paper, we divide the imaging geometry of a photoacoustic computed tomography (PACT) structure that have a circular geometry and have focused transducers of positive or negative focus, into different zones of varying resolution. With using the presented analysis, it is possible to find the maximum range in which the system resolution is at its best value, according to the device specifications. The following equations are used to find the blind spot angles of the system for each source point inside the chamber. These are important for two reasons: the first is that the user can calculate the range in which the imaging system has the best quality by entering the specifications of his PACT system in the equations mentioned in this article. The second is that the user can, depending on his minimum expectations of a PACT system, use the presented equations to calculate the parameters required to design a PACT system.

Keywords— *image resolution, photoacoustic computed tomography, focused transducers, blind spot angles*

I. INTRODUCTION

Photoacoustic imaging is a new imaging technique originally conceived by Alexander Graham Bell [1]. In this imaging method, light is first projected into the target tissue by an optical source. Then the tissue will absorb light and heats up depending on certain conditions. Further heating of the tissue will produce an ultrasound wave [2,3]. Due to its special function, this method has both the advantages of optical imaging and the advantage of ultrasound imaging [4,5]. Since visible light and ultrasound are used in this imaging method, it will not pose a threat to the patient's health [6]. This imaging method has many advantages in terms of price, contrast and resolution compared to other imaging methods such as ultrasound, MRI and CT [7]. Photoacoustic imaging for clinical applications has different planar and cylindrical geometries. the cylindrical geometry is very useful for the diagnosis of breast cancer [8] and Rheumatoid arthritis [9-10] and many other cases [11]. This paper is based on a two-dimensional cylindrical geometry.

In photoacoustic imaging, one way to achieve a high resolution and thus high quality image is to use point-like transducers (point-like detectors). Given that in practice it is not possible to use such transducers, the idea of using the concept of a virtual detector has been proposed in recent years [12]. In this method, a focused transducer is used and the focused location is considered as a virtual detector where, obviously, the size of the focused point is smaller than the size of the transducer element. It is assumed that we have a virtual transducer from the focal point that is propagating or receiving an almost spherical wave whose angular range depends on the numerical aperture (NA) of the transducer [13]. Closed transducer geometries provide a better resolution than other modes. The reason is that in time-reversal reconstructions,

The larger viewing angle of the transducers with respect to a specific photoacoustic source leads to more complete capturing of its spatial frequency components and thus better image quality. Therefore, obviously, the farther we go away from the center of the circle, the lower the quality of the reconstructed points will become. In this case, the number of transducers that can see the desired point will decrease, and some transducers may not even see the point at all. The angular section of the circle where these transducers are located is called the blind angle for that source point.

Based on the presented analysis and equations in this paper for a given system specifications, the value of the blind angle for each source point in the enclosure is determined which is of great importance in evaluating an photoacoustic tomography setup. We have also presented other equations that can be used to determine the largest area in which the system resolution is at its best value. This area is the area where is in the overlap region of the beam of all transducers and therefore, the reconstructed image in this area is in the best possible quality and resolution. All these equations are adjusted in such a way that the user can evaluate the output of a specific photoacoustic imaging system by placing the system specifications such as the radius of the cylinder chamber and the size of the transducer element and their NA, or vice versa according to the expectations from the photoacoustic imaging system.

II. MATERIAL AND METHODS

A. Calculate the maximum quality radius

Fig. 1a shows an outline of a circular chamber with radius R and a positive-focused transducer, and Fig. 1b shows the same outline of a negative-focused transducer. An example source point is also identified to be evaluated in view of angular coverage with transducers. In the following equations, f is the focal length of transducer, and "NA" is the numerical aperture of the transducer. The circle radius is also demonstrated by R . The area of the circle that is visible to all transducers in the photoacoustic imaging region of interest is a circle marked as S_0 in Fig. 1 with radius r_{S_0} as shown in (1). The angle β is shown in Fig. 1. also, the relation of NA is demonstrated in (2) and by substituting in (1), the radius r_{S_0} becomes as presented in (3).

$$r_{S_0} = (R \pm f) * \sin \beta \quad (1)$$

$$NA = \sin \beta \quad (2)$$

$$r_{S_0} = (R \pm f) * NA \quad (3)$$

In (1) and (3), the + sign is for negative focused transducer and the - sign is for positive focused transducer. Therefore, if

the photoacoustic source is in the circle S_0 whose radius is r_{S_0} , it is in the field of view of all transducers and the resolution is at its highest value.

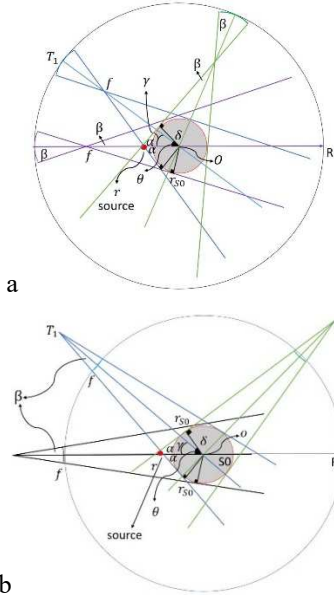


Fig. 1. (a) Outline of the beam edges of several negatively focused transducers. (b) Outline of the beam edges of several positively focused transducers whose focal point is f at the distance from the transducer and the transducers around the cylinder are rotating, marked. A red photoacoustic source is shown in the figure. The range S_0 , which is a circle with radius r_{S_0} , is also shown in the figure.

B. Blind angle calculation for each point source

Next, we proceed to calculate the blind angles of point source located in the chamber. According to Fig. 1, if we consider the red source, for transducers in the upper half of the circle, the first transducer which views the photoacoustic source is the one which its beam bisector is the line passing through the center of the chamber with angle γ with respect to the horizon. And the last transducer that sees the photoacoustic source in the upper half of the chamber is the one whose beam bisector is determined by γ' angle with respect to the horizon, and all the transducers between the two transducers listed above in the upper half of the chamber see the photoacoustic source. Based on the above analysis, it can be concluded that transducers whose beam bisector is not between γ and γ' with respect to the center of the chamber and are located in the upper half of the chamber, do not see the source in Fig. 1, which are called blind transducers. The above analysis is valid for both types of transducers of Fig. 1, i.e. the positive focused mode and the negative focused mode. According to Fig. 1, α is given in (4) and (5), the angle β is given in (6), and angle γ is shown in (7) and (8) according to Fig. 1. The parameter α is also the external angle of triangle $T_1 O r$. By substituting (5) and (6) into (8), we reach to (9), which is the beginning of the blind angle region for the upper half of the cylinder. The end angle of the blind region, which we defined for the upper half of the circle with γ' as shown in Fig. 1, is also given in (10).

$$\sin \alpha = \frac{r_{S_0}}{r} \quad (4)$$

$$\alpha = \sin^{-1} \left(\frac{r_{S_0}}{r} \right) \quad (5)$$

$$\beta = \sin^{-1}(\text{NA}) \quad (6)$$

$$\alpha = \beta + \gamma \quad (7)$$

$$\gamma = \alpha - \beta \quad (8)$$

$$\gamma = \sin^{-1} \left(\frac{r_{S_0}}{r} \right) - \sin^{-1}(\text{NA}) \quad (9)$$

$$\theta + \delta = \gamma' \quad (10)$$

$$180 - \alpha - \beta = \gamma' \quad (11)$$

$$\gamma' = \pi - \sin^{-1} \left(\frac{r_{S_0}}{r} \right) - \sin^{-1}(\text{NA}) \quad (12)$$

Where, r is the distance from the source point to the center of the circle. Therefore, the total angle of the blind region for the source of Fig. 1 for the upper half of the circle, is given in (14), which is obtained by substituting (9) and (12) in (13).

$$\text{upper blind angles} = \gamma' - \gamma \quad (13)$$

$$\text{upper blind angles} = \pi - 2 \sin^{-1} \left(\frac{r_{S_0}}{r} \right) \quad (14)$$

The blind angles for the lower half of the circle are exactly equal to the top half, so the whole angle of the blind region for each source inside the chamber is given in (15), which is exactly twice the blind region for the top half.

$$\text{blind angles} = 2\pi - 4 \sin^{-1} \left(\frac{r_{S_0}}{r} \right) \quad (15)$$

As can be seen in (15), the blind angles for each source inside the chamber depend only on the source distance to the center of the circle and the radius of the circle. From (15) we conclude that the farther we go from the center of the circle, the greater the blind angles for each source would become. Therefore, it can be inferred that the farther we go from the center of the circle, the reconstructed image quality is reduced. The simulation results of this section are given in section III.

In photoacoustic computed tomography (PACT) imaging systems of cylindrical geometry, the object to be imaged is usually placed in the middle of the container, which is another proof of the accuracy of the above analysis.

III. SIMULATION RESULTS

The simulations of this section have been done with k-wave toolbox in MATLAB [14]. In these simulations, we have considered a large number of phantoms as photoacoustic sources. The dimensions of the simulation chamber in each part is different from the other parts and the specifications, number and dimensions of the transducers are also different in each part. In general, several types of positive-focused and negative-focused transducers with different NAs in different dimensions are considered. Also, the bandwidth of transducers and their central frequency according to the commercial standard models are considered 80% and 6.25MHz, respectively. In all simulations, the speed of sound is considered to be constant and equal to 1500 meters per second.

A. maximum quality radius simulations

Our main goal in this section is to find the maximum radius in which the image quality is good and the resolution is high. So in this part, we have to consider the phantoms as small as

possible in order to find the maximum radius of the reconstructed image that the resolution is at its best. For the simulations of this part, a square chamber with a side size of 12 cm, including 2400 by 2400 grids, and each square grid with a side size of 0.05 mm, has been selected. 630 transducers are arranged in a circle with a radius of 54 mm. The focal point of all transducers is 18 mm from the center of the transducer. Three different types of transducers including a positive focused transducer with $NA = 0.2$, a negative focused transducer with $NA = 0.2$, and also another type of negative focused transducer with $NA = 0.4$ were considered. All of these transducers have 80% bandwidth and a central frequency of 6.25MHz.

1) Positive Focused Transducer

Fig. 2a shows a phantom of 4,000 circular photoacoustic sources, each with a radius of 50 microns, or about one-fifth of the transducer center wavelength. Its zoomed image is shown in Fig. 2b. The specifications of the system are given in (16). By substituting (16) in (3) for the positive focused mode, the maximum radius of the imaging system in which the resolution of the reconstructed images is of the highest quality is $r_{s0} = 7.2mm$.

$$\begin{cases} R = 54mm \\ f = 18mm \\ NA = 0.2 \end{cases} \quad (16)$$

In Fig. 2c, according to the presented grids, the center of the chamber is the point (1200 and 1200). If we take a closer look at the radius of 7.2 mm from the center of Fig. 2c shown in Fig. 2d, we will see that the reconstructed photoacoustic sources at distances farther from the 1340 grid, which is the same radius of 7.2 mm from the center, do not have good reconstruction quality were the considered photoacoustic point sources are not fully differentiable. It should be noted that the distance of each source in Fig. 2d from its adjacent source is 0.2mm, which will be 80% of the central wavelength.

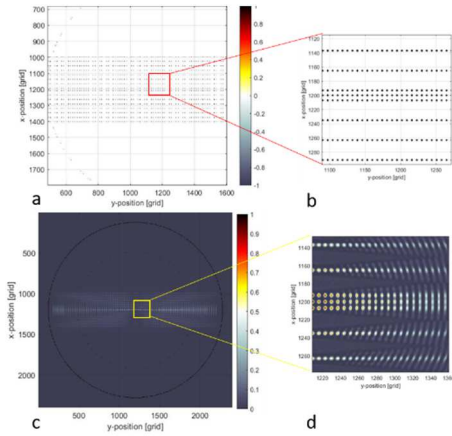


Fig. 2. Maximum quality radius for Positive focused transducer. (a) part of the 4000 initial photoacoustic source point in a circle with a radius of 54 mm. (b) is the part of (a) that is zoomed. (c) reconstructed image. (d) Part of the reconstructed image. As can be seen, from around grid 1340 onwards, the resolution drops significantly compared to before.

2) Negative focused transducer with $NA = 0.2$

In Fig. 3a, unlike the Fig. 2a, 20,000 photoacoustic sources are placed inside the chamber. Its zoomed image can be seen in Fig. 4b. The system specifications are given in (16) with the

difference that these values are for negative focused mode. By substituting (16) in (3) for the negative focused mode, the maximum radius of the imaging system in which the resolution of the reconstructed images is of the highest quality is $r_{s0} = 14.4mm$.

The center of Fig. 3c is the same as Fig. 2c (1200 and 1200). If we take a closer look at the radius of 14.4 mm from the center of Fig. 3c shown in Fig. 3d, we will see that the reconstructed photoacoustic sources farther away from the grid around 1480, or the same radius of 14.4 mm from the center, do not have the best possible reconstruction quality and resolution will not be the best as some points has not been differentiated from each other.

3) Negative focused transducer with $NA = 0.4$

All the specifications of this mode are the same as the previous mode. With the only difference that $NA = 0.4$. The system specifications are given in (17) and by substituting it in (3) for negative focused mode, the maximum radius of the imaging system in which the resolution of the reconstructed images is of the highest quality will be 28.8 mm.

$$\begin{cases} R = 54mm \\ f = 18mm \\ NA = 0.4 \end{cases} \quad (17)$$

The center of Fig. 3e is the same as Fig. 3c (1200 and 1200). If we take a closer look at the radius of 28.8 mm from the center of Fig. 3e shown in Fig. 3f, we will see that the reconstructed photoacoustic sources farther away from the grid around 1740, or the same radius of 28.8 mm from the center, do not have the best possible reconstruction quality and resolution will not be the best as again some photoacoustic point sourced could not be completely resolved from each other.

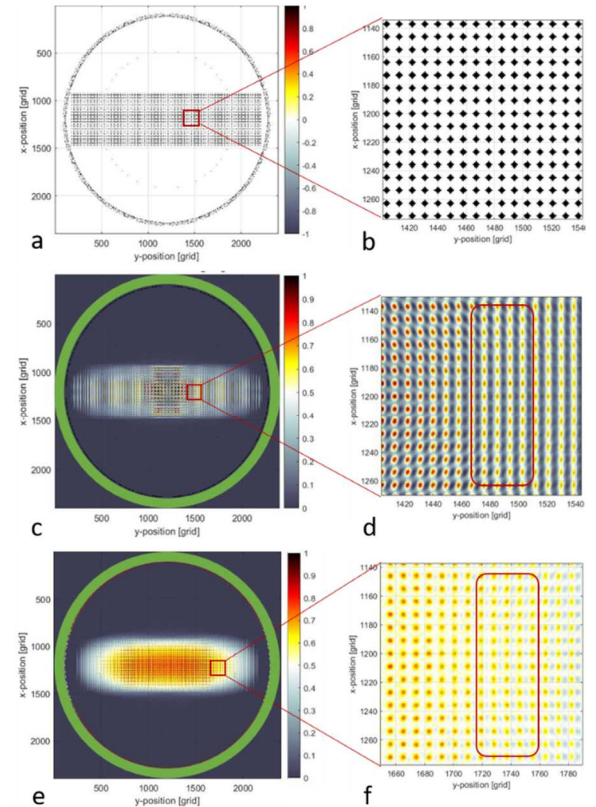


Fig. 3. Maximum quality radius for negative focused transducer. (a) 20,000 photoacoustic source phantoms in a container with a radius of 720 grids or 3.6 cm. (b) zoomed (a). (c) reconstructed with negative focused transducer with NA=0.2. (d) Zoomed image (c). As can be seen, the area in the frame is roughly the boundary between the high resolution part and the low resolution part. (e) reconstructed with negative focused transducer with NA=0.2. (f) Zoomed image (e). As can be seen, the area in the frame is roughly the boundary between the high resolution part and the low resolution part.

B. Acceptance area simulations

In Section II.B, the blind angles for each photoacoustic source were obtained and it was found that each photoacoustic source would have certain blind angles due to its distance from the coordinate center, so the blind angles would be different for each point source. Fig. 4 and Fig. 5 show a number of large photoacoustic sources, each large source consisting of a large number of photoacoustic point sources. Based on the presented analysis in Section II, in the reconstruction of any point source, part of the point source is not reconstructed and has blind angles, so by putting together a large number of point sources, we see what is reconstructed in the output of the image. The amplitude of the package of blind angles that each part has will be different.

In (15) blind angles were obtained for each point, using the blind angle of each point, the value of the ratio of the received angular view area of the point source to the whole area of the point source can be calculated as shown in (18). By substituting (3) in (18) we will reach to (19).

$$acceptance\ area = \frac{2 \sin^{-1}\left(\frac{r_{S0}}{r}\right)}{\pi} \quad (18)$$

$$acceptance\ area = \frac{2 \sin^{-1}\left(\frac{(R \pm f) \cdot NA}{r}\right)}{\pi} \quad (19)$$

In (19) the + sign is for negative focused mode and - sign is for the positive focused mode.

For the simulations of this part, a square chamber with a side length of 8 cm, including 1600 by 1600 grids, and each grid is a square with a side size of 0.05 mm, has been considered. Also, 360 transducers are arranged on a circle with a radius of 720 degrees, equivalent to 36 mm circumference. The focal point of all transducers is 12 mm from the center of the transducer. We chose three different types of transducers, including a positive- focused transducer with NA = 0.2 whose simulation results are shown in Fig. 4, a negative- focused transducer with NA = 0.2 whose simulation results are shown in Fig. 5. All of these transducers have 80% bandwidth and a core frequency of 6.25MHz.

1) Positive focused Transducer:

Fig. 5a shows the Derenzo phantom. The characteristics of the system in this case are given in (20) and by substituting (20) in (19) we obtain (21) where the ratio of the amplitude of the reconstructed points according to their distance from the origin, r , is obtained.

$$\begin{cases} R = 36mm \\ f = 12mm \\ NA = 0.2 \end{cases} \quad (20)$$

$$pos\ acceptance_{area}(r) = \begin{cases} \frac{2 \sin^{-1}\left(\frac{4.8}{r}\right)}{\pi} & . r \geq 4.8 \\ 1 & . r < 4.8 \end{cases} \quad (21)$$

The reconstructed image is shown in Fig. 4b, which consists of several circles with different radii. In each circle, the points closest to the center of the imaging chamber have a higher amplitude than the other points in the circle. This is because the closer we get to the center of the imaging chamber, the smaller the blind area of each photoacoustic source and the larger the field of view, so it will naturally have a higher amplitude. One of the circles is shown in Fig. 4c. The one-dimensional diagram of the A-line shown in Fig. 4c is shown in 4d. The range of A-line points is normalized to its maximum value. Equation (21) for this point is shown in Fig. 4e. The corresponding parts of Fig. 4e and Fig. 4d are plotted together in Fig. 4f. As can be concluded from the comparison of these curves it can be concluded that the result of (21) is in good agreement with simulation results validating the accuracy of the presented equations.

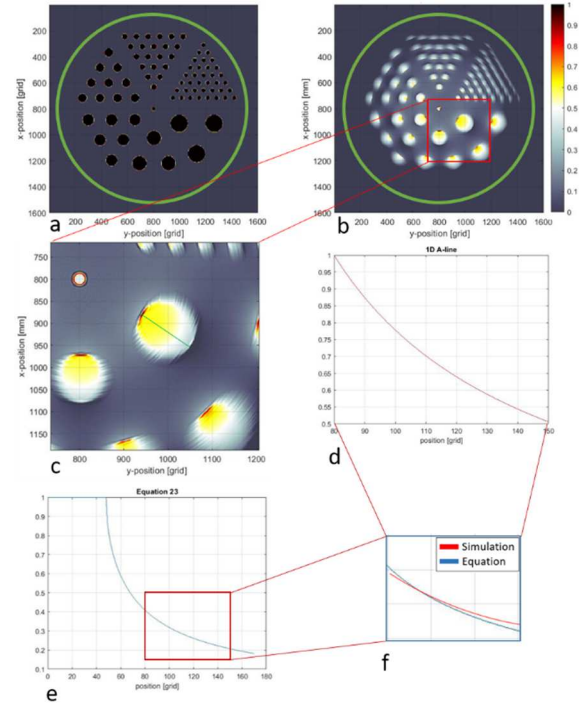


Fig. 4. Acceptance area simulations for positive focused transducer. (a) Derenzo photoacoustic source phantoms in a container with a radius of 720 grids or 3.6 cm. (b) reconstructed with positive focused transducer with NA=0.2. (c) Zoomed image (b). (d) one-dimensional diagram of the A-line shown in (c). (e) Curve (21). (f) Matching (d) and (f) curves.

2) Negative focused Transducer:

Fig. 5a shows the Derenzo phantom. The system specifications, in this case, are the same as in the previous case as presented in (20), except that the transducer, in this case, is negative focus. By substituting (20) in (19) we get to (22) where the ratio of the amplitude of the reconstructed points is obtained according to their distance from the coordinate center which is r .

$$neg_{acceptance}^{area}(r) = \begin{cases} \frac{2 \sin^{-1}(\frac{9.6}{r})}{\pi} & .r \geq 9.6 \\ 1 & .r < 9.6 \end{cases} \quad (22)$$

The reconstructed image is shown in Fig. 5b. As mentioned, in each circle, the points closest to the center of the imaging chamber have a higher amplitude than the other points in that circle. One of the circles is shown in Fig. 5c. The one-dimensional diagram of the line shown in 5c is shown in 5d. Equation (22) is plotted for this point and is shown in Figure 5e. By comparing the corresponding parts of Fig. 5e and 5d as shown in Fig. 5f, it can be concluded that the result of (22) is in good agreement with simulation results again confirming the presented equations for negative focused case.

The slight difference between the two graphs in Fig. 5f and Fig. 4f is due to an error in calculating the beam shape of the transducers in the equations. Achieving the exact boundary of the beam shape of the transducer is impossible because it does not have a definite value and falls with a Gaussian function. As shown in Fig. 1, we set this boundary to a constant value, which is the average of the Gaussian function.

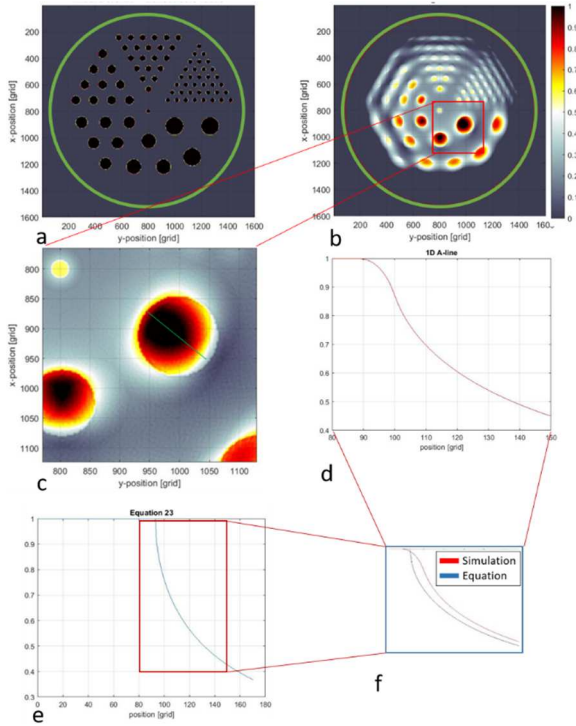


Fig. 5. Acceptance area simulations for positive focused transducer. (a) Derenzo photoacoustic source phantoms in a container with a radius of 720 grids or 3.6 cm. (b) reconstructed with negative focused transducer with NA=0.2. (c) Zoomed image (b). (d) one-dimensional diagram of the A-line shown in (c). (e) Curve (22). (f) Matching (d) and (e) curves.

IV. CONCLUSION

In summary, each photoacoustic imaging system will have different resolution quality depending on its geometry and the characteristics of the transducer used in it. In this paper the transducer used is a focused transducer and we have considered the system geometry to be circular and two-dimensional. Complete equations are presented, with the help of which according to the transducer specifications including

element length and NA and the characteristics of the imaging system such as the radius of the imaging circle, the maximum area in which the imaging resolution is at its maximum is specified. The importance of this is that in many PACT applications the circular target is body imaging such as rheumatoid arthritis diagnosis or many other applications such as breast cancer diagnosis, in which case it is very important that What is the maximum area where the resolution is at its maximum or, on the contrary, it is possible to design the system according to the amount of required area. There are also other equations that can be used to determine what ratio of each point source, depending on its distance from the center of the chamber, is reconstructed with high quality when its image is reconstructed. The area that is not reconstructed is called the blind area, which in addition to the distance from the source to the center, is also highly dependent on the properties of the transducer. This is important because the system can be designed according to an application so that the imaging area used has the most acceptance area. Also, all equations have been validated using various simulations and the results have been compared with the presented equations for their validation.

REFERENCES

- [1] A. G. Bell, "On the production and reproduction of sound by light," *Am. J. Sci.* 118, 305–324 (1880).
- [2] M. Xu, L.V. Wang, "Photoacoustic imaging in biomedicine," *Rev. Sci. Instrum.* 77 (4) (2006) 041101.
- [3] L. V. Wang, "Tutorial on Photoacoustic Microscopy and Computed Tomography," in *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 14, no. 1, pp. 171-179, Jan.-feb. 2008.
- [4] L. V. Wang, "Multiscale photoacoustic microscopy and computed tomography," *Nature Photon.*, vol. 3, no. 9, pp. 503-509, Aug. 2009.
- [5] L. Yao et al., "Photoacoustic computed microscopy," *Sci. Rep.*, vol. 4, May 2014.
- [6] X. Wang et al., "Noninvasive photoacoustic tomography of human peripheral joints toward diagnosis of inflammatory arthritis," *Opt. Lett.*, vol. 32, no. 20, pp. 3002-3004, Oct. 2007.
- [7] Attia ABE et al., "A review of clinical photoacoustic imaging: Current and future trends," *Photoacoustics*, 16:100144, Nov. 2019.
- [8] L. Xi et al., "Evaluation of breast tumor margins in vivo with intraoperative photoacoustic imaging," *Opt. Express*, vol. 20, no. 8, pp. 8726-8731, Apr. 2012.
- [9] L. Xi and H. Jiang, "High resolution three-dimensional photoacoustic imaging of human finger joints in vivo," *Appl. Phys. Lett.*, vol. 107, Aug. 2015.
- [10] Y. Sun et al., "Quantitative three-dimensional photoacoustic tomography of the finger joints: An in vivo study," *J. Biomed. Opt.*, vol. 14, no. 6, Nov. 2009.
- [11] Siyu Liu, Weitao Song, Xinqin Liao, Tony Tae-Hyoung Kim, Yuanjin Zheng, "Development of a Handheld Volumetric Photoacoustic Imaging System With a Central-Holed 2D Matrix Aperture," *Biomedical Engineering IEEE Transactions on*, vol. 67, no. 9, pp. 2482-2489, 2020.
- [12] G. Drozdov and A. Rosenthal, "Analysis of Negatively Focused Ultrasound Detectors in Photoacoustic Tomography," in *IEEE Transactions on Medical Imaging*, vol. 36, no. 1, pp. 301-309, Jan. 2017.
- [13] C. Li, L. V. Wang, "High-NA-based virtual point detectors for photoacoustic imaging," *Proc. SPIE 7177, Photons Plus Ultrasound: Imaging and Sensing 2009*, 71770R Feb. 2009.
- [14] B. E. Treeby, & B. T. Cox, "k-Wave: MATLAB toolbox for the simulation and reconstruction of photoacoustic wave fields". *Journal of Biomedical Optics*, 15(2), 021314 ,2010.